DE.AUTO

CHAPTER 2

THE ENGINE

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

MEMBER EXCLUSIVE • DE.AUTO •

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DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Two: The Engine

“ONE SPARK. FOUR STROKES. ENDLESS SPEED.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

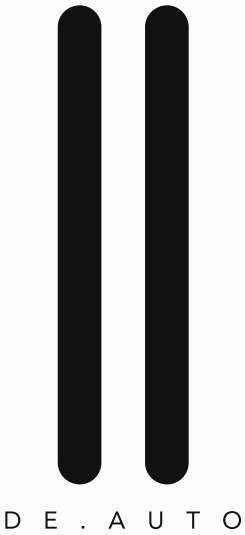
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

DE.AUTO — Because knowledge is power, and cars are passion.

READY?

At 6000 RPM, every cylinder in your engine fires 50 times every second! So why doesn't it melt? And how can a small engine beat a big one? 🔥

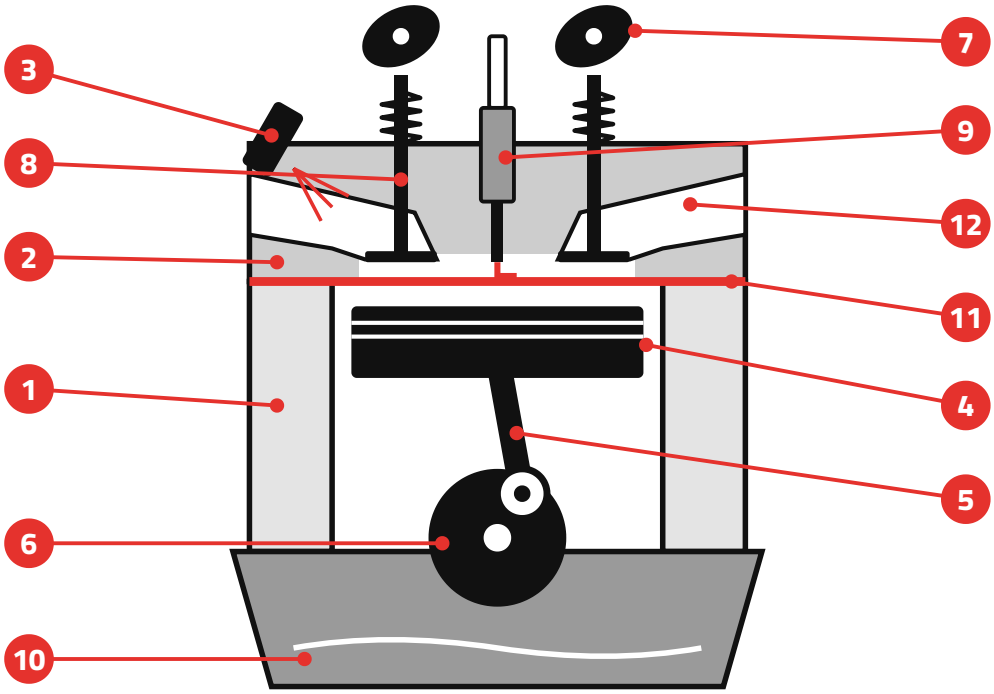
By the end of this chapter you'll know:

- ✓ How an engine works in four steps, and what valves do
- ✓ What engine size means (cc and litres), and whether bigger is always stronger
- ✓ The main engine parts, and the warning signs when they struggle
- ✓ Cast vs forged pistons, and what honing and boring mean
- ✓ What VVT is, and what the throttle and the "fridge" are
- ✓ Engine layouts: inline, V, boxer, W, rotary
- ✓ Horsepower vs torque: who really wins?



Inside the engine: the big picture 🔍

Before we go through each part, here is a simplified map of a slice through one cylinder. Come back to it whenever a new name pops up in this chapter.



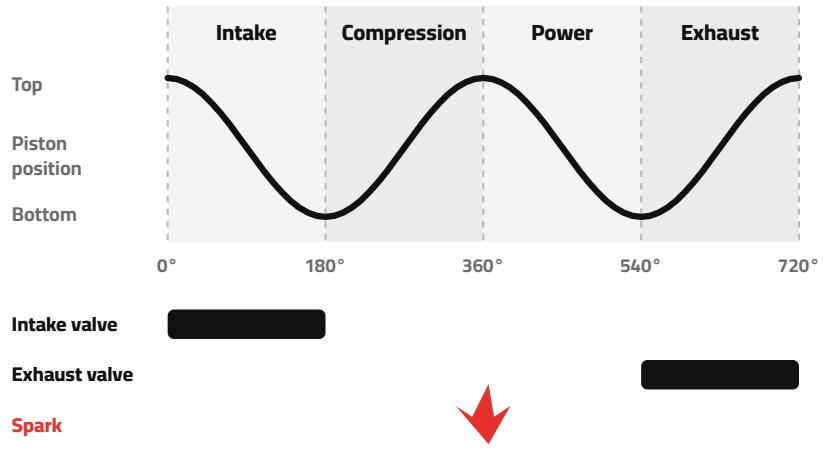
A simplified slice through one cylinder. Real engines have several cylinders and more detail

1 **Block**	2 **Cylinder head**	3 **Injector**
4 **Piston**	5 **Connecting rod**	6 **Crankshaft**
7 **Camshaft**	8 **Valves**	9 **Spark plug**
10 **Oil pan**	11 **Head gasket**	12 **Intake & exhaust ports**

💡 **Did you know?** A multi-cylinder engine is the same slice repeated 3, 4, 6 or 8 times, all working on the same crankshaft. 🤝

How does an engine work? The four-stroke cycle

Almost every petrol car engine runs to the same rhythm: **suck, squeeze, bang, blow**. This time, here it is as a timeline:



Time runs left to right. The crankshaft turns twice (720°) for each full cycle. In real engines the valves open and close with a little overlap, and that is where VVT comes in

1. Intake

The intake valve opens and the piston moves down, so air rushes in. In most engines the fuel comes in with it from the injector. With direct injection, the injector sprays the fuel inside the cylinder itself.

2. Compression

Both valves are shut and the piston rises, squeezing the mixture. The more it is squeezed, the stronger and more efficient the burn.

3. Power

Just before the top, the spark plug fires. The mixture ignites, the gas expands and shoves the piston down. This is where the power is made.

4. Exhaust

The exhaust valve opens and the piston rises, pushing the burnt gases out into the headers (the exhaust manifold).

50 times a second is how often each cylinder fires when the engine turns at 6000 RPM. A cylinder doesn't fire on every turn, only once every two turns of the crankshaft.

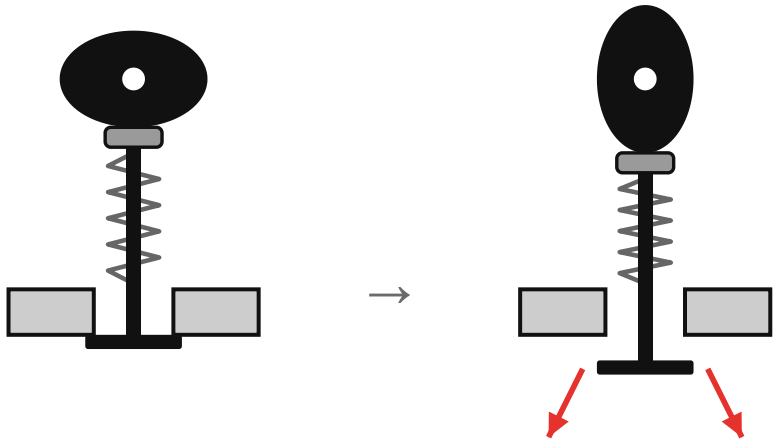
⚠️ **Important correction: Where does octane come in?** In the compression stroke. Octane isn't a measure of petrol "quality". It measures its **resistance to knock** — the mixture detonating at the wrong moment under pressure and heat. We cover it in full in **Chapter 3**.

Valves and camshafts: the engine's gates 🚪

Note: in Gulf garages, valves are called "**takayat**". Same part, just the workshop name. Every cylinder has at least two of these gates:

Intake valve
Opens during the intake stroke and lets the air in (and the fuel mix, in some engines).

Exhaust valve
Opens at the end of the cycle and lets the burnt gases out.



Simplified drawing: the cam pushes the valve down so it opens, and the spring pushes it back shut

Who opens and closes them? The camshaft. It carries egg-shaped bumps called cam lobes. As the shaft turns, a lobe pushes the valve open, and when the lobe moves past, the spring snaps the valve shut. Timing is the secret to power: when it opens, how far it opens, and when it closes.

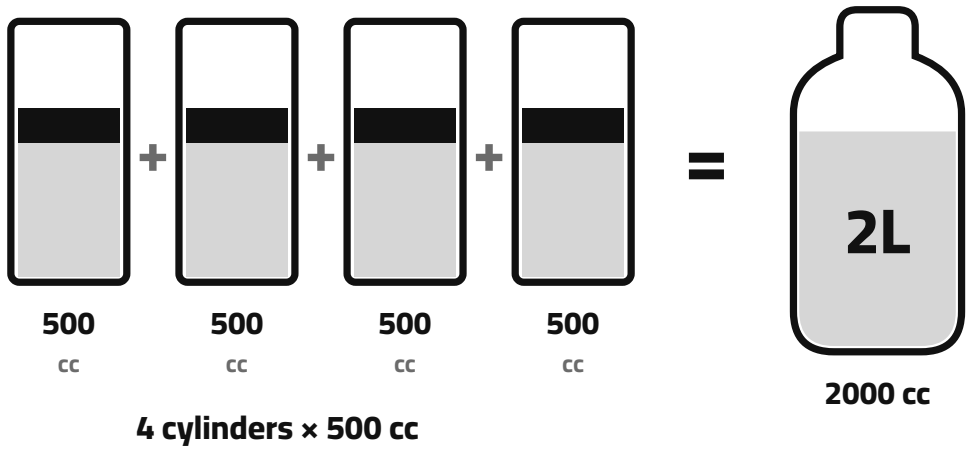
✕ **MYTH**
The engine runs on petrol only. Petrol is the energy, end of story!

✓ **FACT**
Petrol on its own doesn't burn. An engine needs **air (oxygen), fuel, compression, a spark and correct timing** all at once. Miss one and the engine struggles or stops.

💡 **Did you know?** Many modern engines have **4 valves per cylinder** (2 intake and 2 exhaust) so they breathe better. Diesels have no spark plugs: the high air pressure heats the air so much that the diesel ignites by itself.

Displacement: what does 2.0 litres mean?

Engine size (Displacement) is the total volume the pistons sweep through in all the cylinders. It is measured in cubic centimetres (CC) or in litres (L).



Displacement = the combined volume of all the cylinders. Example: four cylinders = a 2.0-litre engine

1000 CC = one litre (1L). So a 1600 cc engine is the same as a 1.6-litre engine.

The general rule: the bigger the displacement, the more air and fuel the engine can burn, so the potential power rises. That is why a 1.6-litre economy family car is a different animal from a 6.2-litre V8 sports car, and the latter usually drinks more fuel.

✕ MYTH
More cc means more power, full stop.

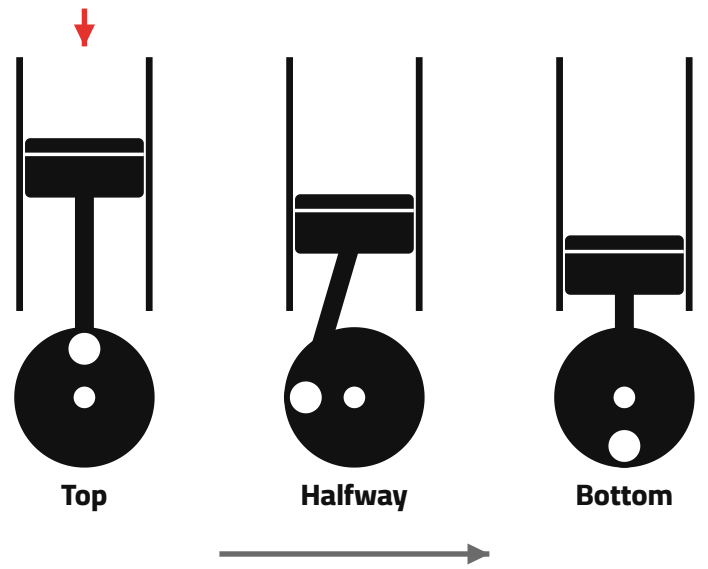
✓ FACT
Displacement is only part of the story. Design and technology matter a lot: small engines with a **turbo** (an air booster) push in more air and compete with bigger engines. We cover turbo details in later chapters.

🎯 Challenge 1 **A 4-cylinder engine where each cylinder is 400 cc. What is the engine size?**

A) 1.2 litres B) 1.6 litres C) 4.0 litres

The skeleton: from combustion to rotation

The explosion pushes the piston up and down. Turning that motion into rotation is the job of this team:



The piston moves up and down, and the crankshaft turns that motion into rotation

Engine block

The solid heart (iron or aluminium) that holds the cylinders. Most other parts bolt onto it.

Cylinder head

The lid on top of the block. It holds the valves, the spark plugs and the air, oil and coolant passages.

Pistons (Rings)

Metal parts that slide up and down inside the cylinder. Rings around them seal in the pressure and control the oil.

Connecting rods

They link each piston to the crankshaft and pass on the push.

Crankshaft

Turns the up-and-down motion into rotation, which then goes to the gearbox and on to the tires.

Flywheel

A heavy disc on the end of the crankshaft that keeps the rotation smooth. The starter spins it to start the engine.

> 💡 **Did you know?** The crankshaft turns two full revolutions for every full combustion cycle (720°). In a multi-cylinder engine, each piston works at a different moment from the others so the rotation stays smooth.

Pistons: cast or forged? 🔨

Most cars leave the factory with **cast** pistons. **Forged** ones come up a lot in performance talk. What is the difference?



Cast

Metal is poured into a mould

Forged

Metal is pressed and shaped under force

Approximate drawing. The dots and lines hint at the metal's inner structure; it is not a real photo

	Cast	Forged
How it is made	Molten metal poured into a mould	A piece of metal hammered or pressed into shape
Strength	Fine for normal everyday use	Usually stronger, and handles higher pressure and load
Heat expansion	Usually less, so it runs quieter	Usually more, so it needs looser clearances and can sound louder when the engine is cold
Price	Cheaper and more common	More expensive
Where it helps	Daily-driver cars	High-boost and high-RPM engines

✕ MYTH

A forged piston is better than a cast one in every case.

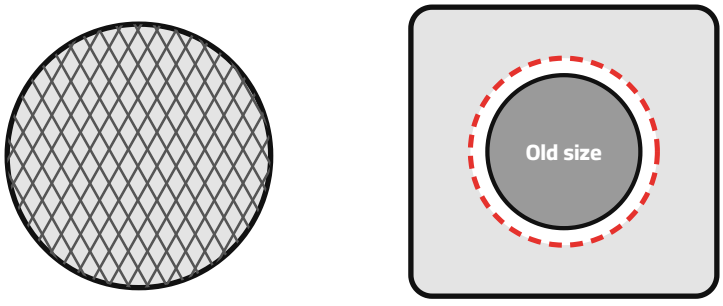
✓ FACT

Not always. Cast pistons are enough for, and suit, most stock engines. Forged ones matter when pressure, heat and RPM go beyond what the factory designed for, and they have to be fitted properly to precise tolerances.

🧩 **Important:** The piston type is tied to the rest of the engine: the block, rings, cooling and tuning. A strong part on its own doesn't make a strong engine.

Honing and boring: why do they open up the engine? 🛠️

When an engine comes out for a rebuild, you will hear two words from the workshop: "**honing**" and "**boring**". Here is what they mean:



Honing: a fine crosshatch
Holds oil on the cylinder wall

Boring: a wider cylinder
(The red circle = the new size)

Simplified illustration

Honing
Smoothing the cylinder wall with a special tool, leaving a fine crosshatch pattern. That pattern holds a film of oil so the rings seat properly and slide smoothly.

Boring
Cutting the cylinder out to a larger diameter with a boring machine, when the wall is worn or scratched. After that you need **bigger** (oversize) pistons to match the enlargement.

Oversize
A piston bigger by an amount set by the manufacturer, for example half a millimetre more or less, depending on the engine.

The cylinder after a rebuild
New pistons and rings go in, and the clearances are checked precisely before assembly.

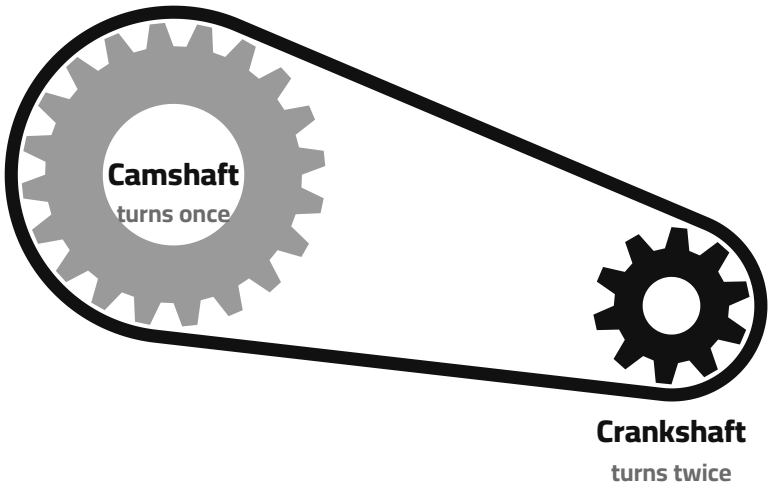
How a rebuild goes (short version)

1. The engine is taken apart, and the parts are cleaned and measured.
2. A decision is made: is honing enough, or is the wall worn and in need of boring?
3. After boring comes honing, to bring back the pattern on the wall.
4. New pistons, rings, bearings and a head gasket go in.
5. The engine is started carefully, on the right oil, for the break-in period as the workshop recommends.

⚠️ **Heads up:** Boring can't go on forever: the more you open it up, the thinner the cylinder wall gets. Professional shops work out that limit before they start.

The timing belt: the thread that holds the whole engine together ⛓️

The crankshaft is at the bottom and the camshafts are at the top. What connects them and keeps them working in the right timing is the **timing belt** or the **timing chain**.



Simplified drawing: the bigger gear turns at half the speed of the smaller one

2 : 1 The crankshaft turns twice for every one turn of the camshaft. The reason: each valve opens only once per full combustion cycle (two turns of the crankshaft).

Belt

Reinforced rubber, quiet and light. It has a replacement interval in the service manual, and you must stick to it.

Chain

Metal, and it usually lasts longer. But it stretches over time and makes noise when it is tired.

❌ **Putting off the belt change past its due date.** If it snaps on an “interference” engine, the valves hit the pistons and the repair bill is huge. Your owner’s manual says which type your engine is.

🎯 **Challenge 2** **Why does the camshaft turn at half the speed of the crankshaft?**

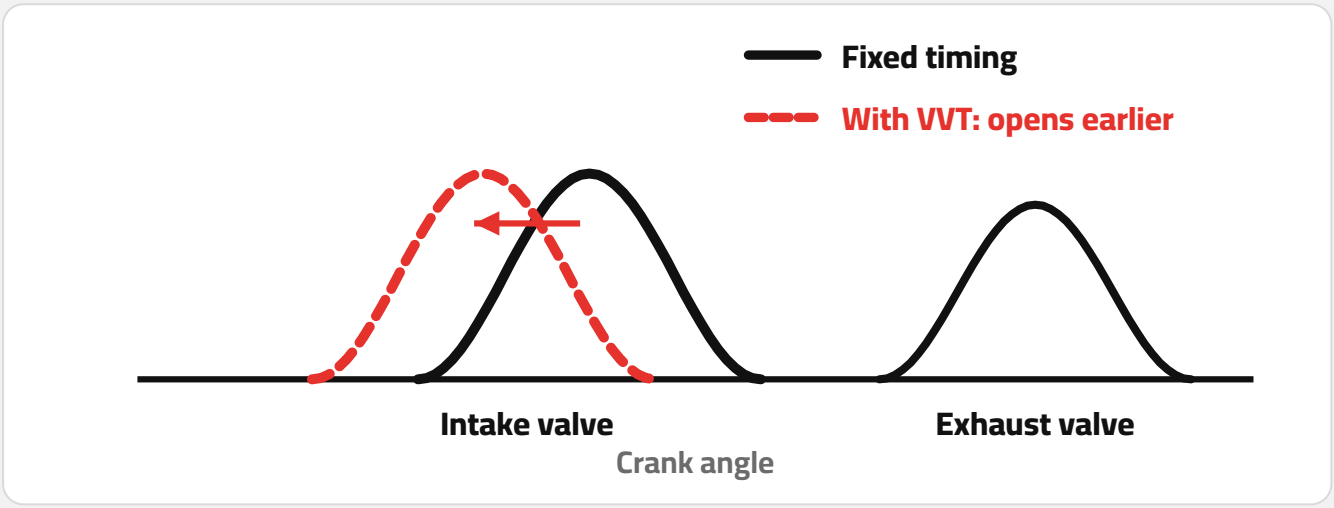
A) To save fuel

B) Because each valve opens once every two turns of the crankshaft

C) Because the belt slips

VVT: the engine that changes its own timing 🧠

Imagine an engine whose valves open at the same moment in every condition. Timing that suits high RPM can make the engine rough at low RPM, and the other way round. The fix? **variable valve timing**, generally known as **VVT**.



The curve shows when the valve is open. VVT slides the curve along without changing its shape, depending on RPM and load

How does it work? (the short version)

On the end of the camshaft there is a part called a Cam Phaser. The computer sends it engine oil pressure, which twists the camshaft by an angle relative to the crankshaft, so the valve opening comes earlier or later depending on RPM and load.

Better torque
Across a wider RPM range, not just at the top.

Lower consumption
The engine breathes more efficiently for each situation.

Cleaner exhaust
Finer control over how much burnt gas stays in the cylinder.

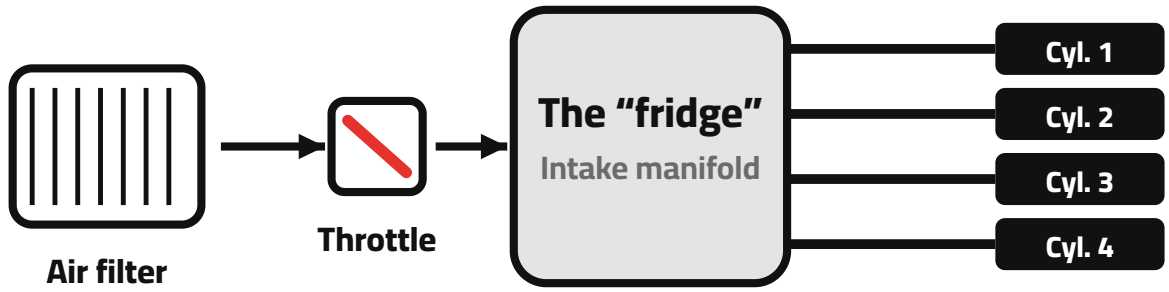
Every brand has its own name
A trademark name for roughly the same idea. The name differs, but the general idea is close.

💡 **Did you know?** Some systems also change the valve **lift** (how far it opens), not just the timing.

❌ **Neglecting the oil.** Most VVT systems run on engine oil pressure. Old, low or wrong oil can affect how they work. Stick to the oil type and change interval in your owner's manual.

The air's journey: the throttle and the "fridge" 🌬️

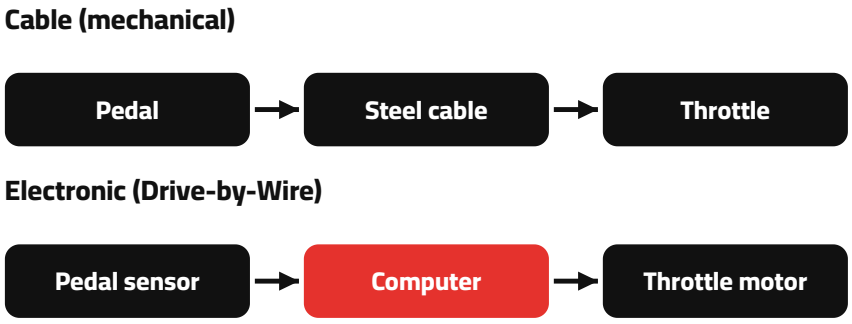
The engine "breathes" air, and the better its path is set up, the better it performs. These are the stops along the way:



The air's journey from the filter to the cylinders

Throttle body

A disc-shaped valve that rotates inside the pipe, opening and closing to control how much air gets in. The more it opens, the more air comes in, the more fuel you can burn, and the more power you get. Two types:



Cable: Your foot pulls a cable attached straight to the throttle. Simple and old-school.

Electronic (Drive-by-Wire): The pedal sends a signal to the computer, and the computer decides the throttle angle and moves it with a motor. This helps with traction control, cruise control and driving comfort.

Intake manifold — also known as "the fridge"

In the Gulf you will hear it called "**the fridge**". That is the garage nickname we like to use, and the proper name is the Intake Manifold. Its job is to share the air out evenly between all the cylinders. In some boosted engines, a built-in air cooler (Intercooler) sits inside it.

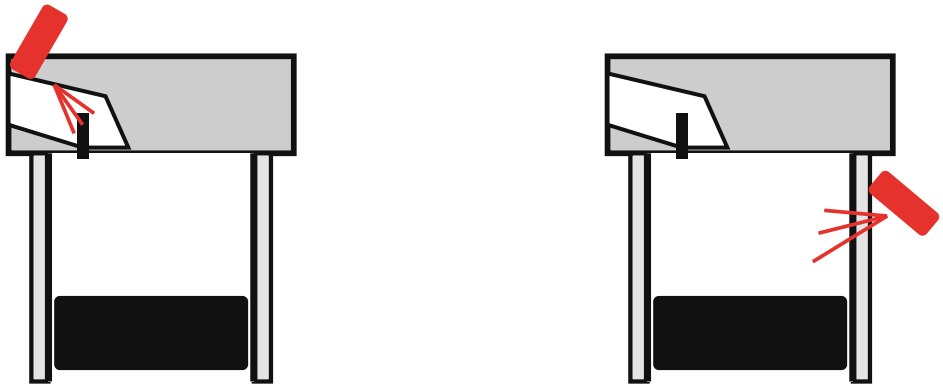
Spark, injector and cover

Spark plugs

They fire the spark that ignites the mixture in petrol engines. They wear out over time, so change them on the schedule and type given in the owner's manual.

Fuel injectors

They spray petrol in a precise amount under computer control, forming a fine mist that burns faster and cleaner.



Port injection
Before the intake valve

Direct injection
Inside the cylinder

Simplified comparison drawing. The red part is the injector and its spray

Port injection: The injector sprays before the intake valve, and the fuel enters along with the air.

Direct injection: The injector sits inside the cylinder itself and sprays at high pressure. Some engines combine both types.

Valve cover

A cover on top of the head that protects the valves and cams and holds the oil in. It houses a valve called the PCV, which pulls vapours that leak past the pistons into the crankcase and sends them back to be burned, instead of venting them to the air. Between the cover and the head sits the **valve cover gasket**.

❌ **An oil leak from the cover that gets ignored.** The sign: oil around the cover, a burning-oil smell, or light smoke from under the hood when oil drips onto hot parts. Often just replacing the gasket fixes it, but do not put it off.

💡 **Did you know?** Where the fuel and air get mixed: in direct injection it happens inside the cylinder itself, and in port injection it happens in the port before the valve. The old carburettor did it in a single spot. Details are in **Chapter 3**.

Oil and the dipstick: the cheapest insurance for your engine 🛢️

Engine oil does more than just “lubricate”. Here are its jobs:

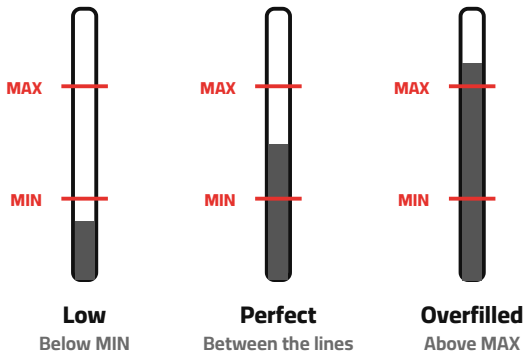
Reduces friction
A film of oil keeps moving metal parts apart.

Cools
Soaks up heat from parts the coolant can’t reach.

Cleans
Carries dirt to the filter.

Seals and protects
Helps the rings seal in pressure, and protects against rust.

Dipstick: check it yourself



How the oil level shows on the dipstick

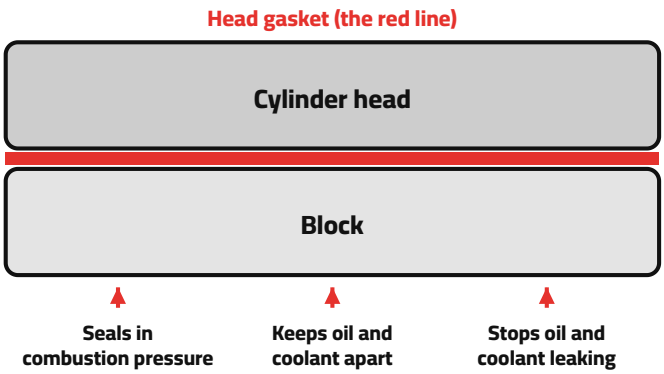
1. Park on level ground, switch the engine off, and wait a few minutes for the oil to drain back.
2. Pull out the dipstick and wipe it with a clean cloth.
3. Push it all the way back in, pull it out again and read the level.
4. It should sit between MIN and MAX.

❌ **Ignoring the oil level.** Too little harms the engine, and too much, above MAX, causes problems as well. Some modern cars have no dipstick, and you read the level from the instrument cluster as the owner’s manual says.

🎯 Challenge 3 **Where should the oil level be on the dipstick?**

A) Below MIN is fine | B) Between MIN and MAX | C) Above MAX is better

Head gasket and water pump



The gasket between the head and the block has three jobs

Signs of a blown head gasket

🚨 The engine overheats.

🚨 Coolant keeps dropping with no obvious leak, or there are bubbles in the coolant tank.

🚨 Milky oil (it looks like mayonnaise), or oil in the coolant.

🚨 Thick white smoke from the exhaust at times (water vapour).

🚨 Generally weak performance.

✅ **DE.AUTO tip:** To prevent damage: keep the cooling system healthy, and don't keep driving once the engine is overheating.

Water pump

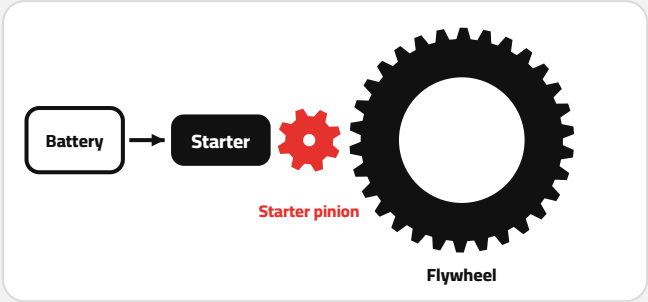
It pumps coolant through the engine and the radiator. On some engines it is driven by the timing belt, which is why it gets replaced along with the belt according to the manual. Signs: **temperature suddenly rising, a coolant leak near the pump, a strange noise from the belt**.

❌ **Ignoring the temperature gauge.** If the needle goes into the red, pull over somewhere safe and switch off the engine. Carrying on can wreck the gasket and the head.

❌ **Opening the radiator cap while the engine is hot.** Steam and pressurised coolant can burn you. Wait until it has cooled down completely.

Starter and rod bearings

Starter motor



The starter pushes its small gear against the flywheel to turn the engine over

An electric motor that draws power from the battery and pushes its small gear against the flywheel so the engine turns over. Once the engine runs, the gear pulls back.

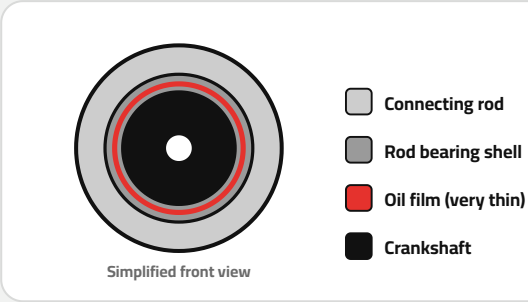
Signs it is struggling:

🚨 A “click” and no cranking.

🚨 Very slow cranking.

🚨 Starts sometimes and not others.

Rod bearings



The oil film is what keeps metal from touching metal

Thin metal shells between the connecting rod and the crankshaft. Thanks to a very thin oil film between them and the crank, they reduce friction and let the engine turn smoothly.

Signs it is struggling:

🚨 A deep knocking sound from the engine.

🚨 Dropping oil pressure.

🚨 Metal particles in the oil.

✅ **DE.AUTO tip:** Starter: if it is weak, check the battery and the electrical connections before replacing it. Many “dead starters” turn out to be a weak battery.

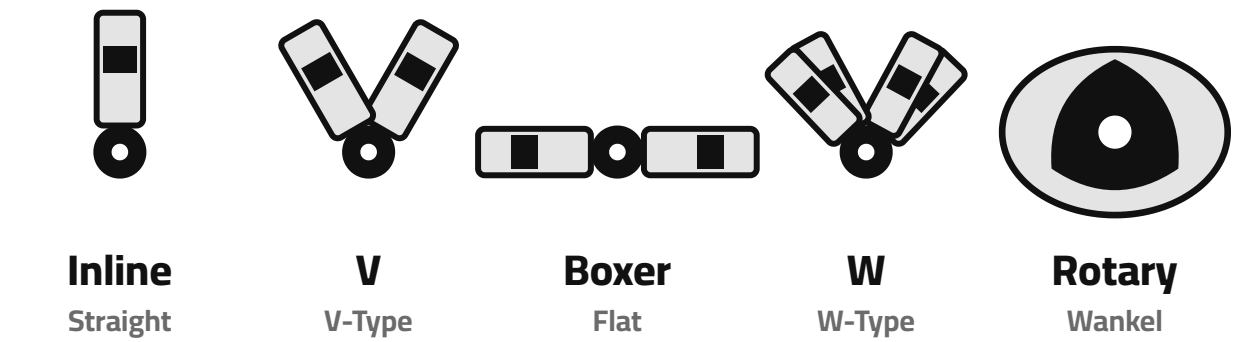
✅ **DE.AUTO tip:** Rod bearings: prevention means the right oil, regular changes, warming the engine up before high RPM, and checking the oil level often.

🗣️ **Among enthusiasts:** Some high-revving engines need extra attention to bearing condition, which is why some buyers check them before buying. That is not a verdict on any particular type of engine, since the actual condition depends on maintenance and use.

❌ **Running a cold engine at high RPM.** The oil has not reached every part yet, which adds wear on the bearings and other parts.

Engine layouts: how are the cylinders arranged?

The idea is simple: the same four strokes, only the cylinder arrangement changes. Each layout has its pros and cons:



Front view of the cylinder layouts. A simplified schematic

Inline

The cylinders sit in a straight row, like 4 and 6 cylinders. A simple design that is usually easier to service. The inline-6 is naturally smooth. Its downside is length as cylinders are added.

V engine

Two rows of cylinders at an angle in a V shape over the same crankshaft. It shortens the engine, so 6, 8 and 12 cylinders fit in a smaller space.

Boxer (Flat)

The cylinders lie flat and face each other, as if the pistons were “punching” one another. A low centre of gravity and good stability. The downsides are width and difficult access to some parts.

W engine

Basically two narrow V rows side by side. It packs many cylinders (like 12 and 16) into a reasonable space. Complex and costly to maintain.

💡 **Did you know?** The inline-6 is naturally smooth because the vibrations of its cylinders cancel each other out. The boxer layout has been used for decades by some manufacturers, and each manufacturer picks the arrangement that fits the philosophy of its cars.

✕ MYTH

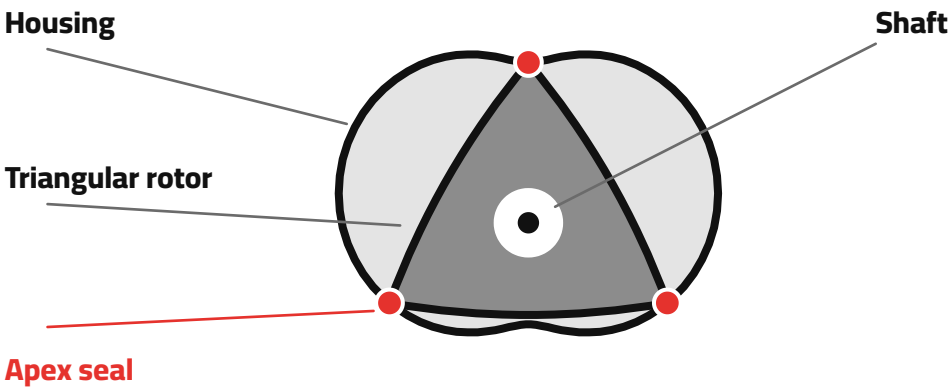
A V engine makes more torque because it is “shorter and wider”.

✓ FACT

It is not the shape. The V is useful because it is **compact and short**, suits many cylinders and runs smoothly. Torque is set by **engine size, design, tuning and boosting** (turbo or supercharger), not by the letter V.

The rotary engine: the beautiful oddball 🌀

Most engines have pistons that go up and down. But there is another design called the **rotary** (Rotary / Wankel): instead of pistons, a **triangular rotor** spins inside a “peanut”-shaped housing, and each side of the triangle does the same four strokes: intake, compression, combustion, exhaust.



Very simplified. Real engines also have intake and exhaust ports, spark plugs and extra seals

✅ Pros

Small and light for its power, very smooth running, and no conventional valves or pistons going back and forth.

⚠️ Sealing

Apex Seals hold in the pressure between the rotor and the housing. They wear over time, and that is one of its best-known weak points.

⚠️ Oil

By design it injects a small amount of oil into the combustion to lubricate the seals. So oil consumption is part of its nature, and you should keep an eye on the level.

⚠️ Fuel and upkeep

It usually uses more fuel than a piston engine of the same power, and needs closer attention.

💡 **Good to know:** Some manufacturers used it in older models. Its proper name is the “rotary” or “Wankel”, after its inventor.

🎯 Challenge 4 **What is the main reason a V engine is useful in cars?**

A) It always makes more torque than an inline

B) Compact and short for a high cylinder count

C) It doesn’t need oil

Horsepower and torque: who wins? 🏁

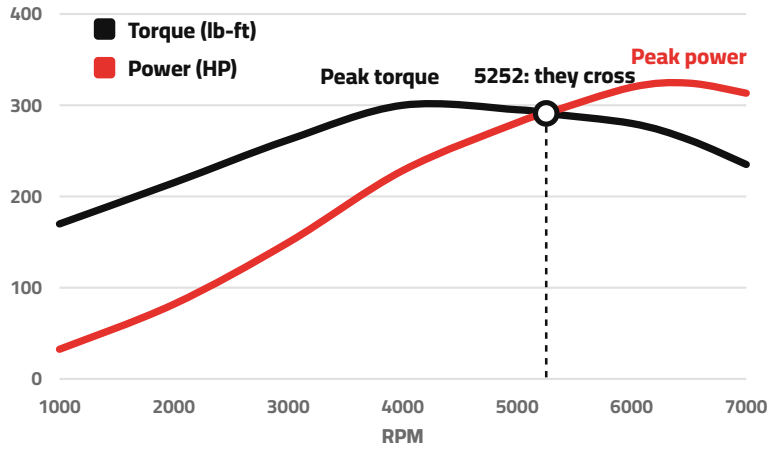
Ever heard someone say “my car has 500 horsepower” and someone else say “it makes 800 newton-metres”? Those two sentences describe different things, but they are more closely linked than you would expect.

Torque
The twisting force on the crankshaft. It is like pushing on a wrench: how much can you twist? Measured in Nm or lb-ft.

Horsepower (power)
How fast the work gets done. Same force, but **how many times per second** can you repeat it? Measured in HP or kW.

HP = Torque (lb-ft) × RPM ÷ 5252

kW = Torque (Nm) × RPM ÷ 9549



A simplified, hypothetical example of a naturally aspirated petrol engine, not any real car. Power is calculated from torque using the formula

Torque peaks roughly in the middle of the range, then falls away.

Power keeps climbing even after torque falls, because RPM rises faster than torque drops.

At 5252 RPM exactly, the two numbers are equal in lb-ft and HP units.

5252 RPM

The number where the torque (lb-ft) and power (HP) lines cross on any chart. Below it, power is numerically lower than torque, and above it, higher. It is not magic, just the result of the formula.

So which matters more? The answer: both!

Let's understand it with an example:

Two people carrying bags
The first lifts one heavy bag and walks slowly, the second lifts half the weight but walks twice as fast. Both move **the same weight per minute**. Torque is "how much you lift", and power is "how much you move per minute".

Why does torque matter?
It is what gives you that "push" from low down, especially with a heavy load or on climbs.

Why does horsepower matter?
It shows how well the engine keeps pulling hard at high RPM, and it usually sets the top-end performance.

And the gearbox?
Gears multiply torque before it reaches the tires. A high-revving engine with less torque can accelerate as well as a higher-torque engine if the gearbox gives it the right gearing.

✕ MYTH
Torque is the "real power" and horsepower is just ability at high RPM.

✓ FACT
Both are **real power**. Power is calculated from torque and RPM, and you can't separate them. What counts is the whole curve of the engine: where its torque peaks and where its power peaks.

💡 **Did you know?** The famous line "horsepower sells cars, torque wins races" is attributed to Carroll Shelby. It is a nice one to drop in conversation, but it is a neat shorthand rather than a law of physics.

✅ **DE.AUTO tip:** If you are comparing two cars, don't look at just one number. Look at **the RPM where the torque peaks**, the weight and the gearbox, and test drive legally and safely.

❌ **Judging a car by one number.** "500 horsepower" without weight, gearbox and RPM doesn't give you the full picture. Numbers are read together.

🎯 Challenge 5 **An engine makes 175 lb-ft at 3000 RPM. Roughly how many horsepower is that?**

A) About 50 B) About 100 C) About 200

What controls engine power? 💪

An engine is like a lung: the more air you take in and burn properly, the more power you get. Here are **six general ideas** that performance people mention. We list them here just to help you see the big picture:

1. A bigger or newer engine design
More displacement, more cylinders or a better design. That is decided at the factory; it is not a simple "modification".

2. Better breathing
Camshafts with different timing, or VVT systems, change how the engine breathes.

3. Tuning
Changing the computer's commands: fuel, spark and more. It may affect your warranty and your compliance with emissions rules.

4. Forced induction
A turbocharger or supercharger pushes in more air. It increases the load on the parts.

5. A freer-flowing exhaust
Headers (the manifold), a downpipe and a cat-back system. Some of these touch the **catalytic converter (the "cat")**, and removing or disabling it is **against the regulations**.

6. Fuel type
Higher octane isn't automatically "extra power". It resists knock, and helps engines designed for it. Details are in Chapter 3.

🛑 **DE.AUTO warning: Important:** These are general ideas for knowledge only, not an invitation to try them and not a promise of any gain in performance. Any real modification is subject to local regulations, may affect the warranty, safety and periodic inspection, and must be done by certified specialists. (We will learn more about modification parts through the chapters of the series.)

✕ MYTH
Higher-octane petrol makes any car more powerful.

✓ FACT
Usually not. High octane prevents knock, which helps if your engine is designed or tuned for it. Use what your owner's manual recommends.

➡️ Why is 91 petrol different from 95? How do air and fuel get in so precisely? What about injectors and carburettors? All of this is in **Chapter 3: Fuel and Air Systems**. Without the right fuel and air, an engine is just a lump of metal. 🚗💨

Garage talk vs proper names 📖

Step into a Gulf workshop and you will hear some unfamiliar words. Here is your dictionary:

Gulf garage talk	Proper name	What it means
Takayat	Valves	Valves (intake and exhaust)
Thalaja ("the fridge")	Intake Manifold	Intake manifold, shares the air out between the cylinders
Throttle bodi	Throttle Body	Throttle body, controls how much air comes in
Sbayek	Rod Bearings	The bearings between the connecting rod and the crankshaft
Jilda ("skin")	Gasket	A gasket that stops leaks (like the valve cover gasket)
Hed gazget	Head Gasket	The head gasket, between the head and the block
Krank	Crankshaft	The shaft that turns piston motion into rotation
Kamat	Camshaft	The camshaft, opens the valves
Basatim / Ringat	Pistons / Rings	The pistons and their rings
Taiming	Timing Belt / Chain	The belt or chain that keeps the crankshaft and camshafts in sync
Self	Starter Motor	The starter motor
Tarumba	Water Pump	The water pump
Hadafa	Flywheel	A heavy disc on the crankshaft
Injectorat	Fuel Injectors	Fuel injectors
Mi'yar	Dipstick	The oil-level stick
Honing / Tahween	Honing	Smoothing the cylinder wall and giving it its pattern
Boring	Boring	Opening the cylinder out to a bigger size
Dubba ("the bear")	Catalytic Converter	The catalytic converter, cleans the exhaust
Headerz	Exhaust Manifold / Headers	The exhaust manifold
Ahsina / Torque	HP / Torque	Power / torque

Wrap-up 📌

Now you get how an engine breathes and works: four strokes, valves and cams, displacement, horsepower and torque, and parts whose warning signs you know. You also understand cast and forged pistons, honing and boring, VVT, and the throttle and the "fridge". You can talk shop in the garage and know what you are saying! 🔥

🎯 Challenge answers:

1) 1.6 litres (4 × 400 = 1600 cc).

2) Because each valve opens once every two turns of the crankshaft.

3) Between MIN and MAX.

4) Compact and short for a high cylinder count.

5) About 100 horsepower (175 × 3000 ÷ 5252 ≈ 100).

➡️ Chapter 3: **Fuel and Air Systems**. We will learn the difference between 91 and 95, injectors and carburettors, and how air and fuel affect power. Get ready! 🚀